Mid Sem Exam –I, Oct 2023

Class: B.Tech I[st] Sem. Branch: CSE-A Code: CHB-101 A (Applied Chemistry)

Date: 30.10.23 Room: 105/216 MM 10 Time: 11.00 PM to 12.00 PM

Note: Attempt any two questions (1 question from each unit).

[Bloom's Taxonomy: BL-1: Remember, BL-2: Understand, BL-3: Apply

Q. 1. (a) Write principle of ion exchange process of water softening with diagram. (b) Write short note on caustic embrittlement. [2.5+2.5=5]

CO: 1	BL: 1/2

Q. 2. (a) What is Calgon conditioning? How it is superior over phosphate and carbonate conditioning. (b) Calculate the amount of lime required for softening of 50000 L of hard water containing $Ca(HCO_3)_2$ = 243 mg/l, $Mg(HCO_3)_2$ = 73 mg/l, $CaSO_4$ = 102 mg/l, $MgCl_2$ = 95 mg/l, NaCl = 428 mg/l and urea = 139 mg/l. Purity of lime is 94% and that of soda is 86.0%. [2+3=5]

CO: 1	BL: 1/3

Q. 3. (a) What reaction will take place if Zn rod is dipped in $CuSO_4$ soution and Cu rod dipped in AgCl solution? How will you detect the presence of $ZnSO_4$ and $CuCl_2$ in the solution? (b) Define electrode potential and its type [3+2=5]

CO: 2	BL: 2/1

Q. 4. Discuss Electrochemical cell and its type with well labelled diagram. [5]

CO: 2	BL: 1/2

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Mid Sem Exam –II (CSE'A') February 2023

CHB 101 (Applied Chemistry) MM 10

Date: 13.02.23 (Monday) Time:10.30-11.30 am Room: 103, 105

Note: Attempt any 2 questions.

Q.1 Define electrode potential. Draw a well labeled diagram of electrochemical cell having Cu and Zn electrodes. Write cell notation for this cell. 5

Q.2 Define corrosion. Discuss types and disadvantages of corrosion. 5

Q.3 Define Polymers. Write classification of polymers on various basis with specific examples. What are conducting polymers? 5

Q.4 Write short notes on: (any two) 5

(i) Nano materials

(ii) Li Ion Battery

(iii) Prevention of corrosion

(iv) Nernst's equation

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Mid Semester Test I–Jan 2023

Branch:	**AIADS**		
Semester	**I Sem**	**Max. Marks-20**	
Subject:	**Digital Electronics (CSA102)**	**Time: 1Hrs**	
Note:	**3. Attempt any two parts from question 1 and 2. 2. Attempt one part from question 3.**		

S.No		Question Marks		CO Att.
1	(A)	i) Convert the hexadecimal number F3A.7C2 to binary and octal. ii) Multiply the following numbers without converting to decimal. (a) $(367)_8$, and $(715)_8$.	3	
	(B)	Subtract following number using vii) 6342 - 3151 using 10's compliment viii) 10110 – 1010 using 2's compliment ix) 3145 - 2321 using 9's compliment		
	(C)	Represent decimal number 6542 and 3124 in (a) BCD, (b) excess-3 code, (c)2421 code (d) Gray code		
2	(A)	Write De-Morgan's Theorem and find compliment of following using it (e) $(x + y' + z)\ (x' + z')\ (x + y)$ (f) $((A+BC')'+D(E+F')')'$	8	
	(B)	Convert the following to the other canonical fonn: (a) $F(x, y, z) = \Sigma(1, 3, 7)$ (b) $F(A, B, C, D) = \pi(0, 1,2,3,4,6,12)$		
	(C)	Simplify the following Boolean expressions using four-variable maps: (A) $B'D + A'BC' + AB'C + ABC'$ (B) $AB'C + B'C'D' + BCD + ACD' + A'B'C + A'BC'D$		
3	(A)	Draw a Full adder circuit with its truth table and construct a full adder using half adder circuit.	4	
	(B)	Explain the 4x1 multiplexer with truth table and draw its circuit diagram.		

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